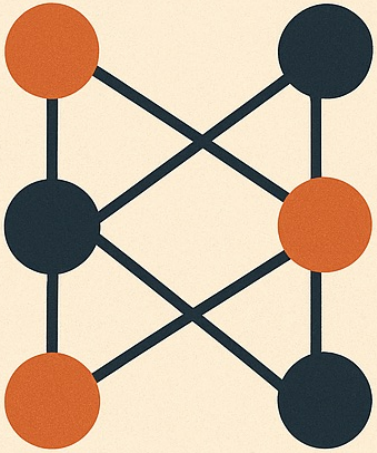
INTRO TO MACHINE LEARNING

AND NEURAL NETWORKS





Lecture 3: Linear Regression - Part II

#Agenda

1) Linear Regression → Feature Evaluation

2) Code from scratch

Recap

1) Cars 24 Data ⇒ { x-feature : age, model, make ...
y-feature : selling price.
target

2) EDA

3) Data preprocessing ⇒ { Categorical Encoding → Target Encoding
Normalization / Std

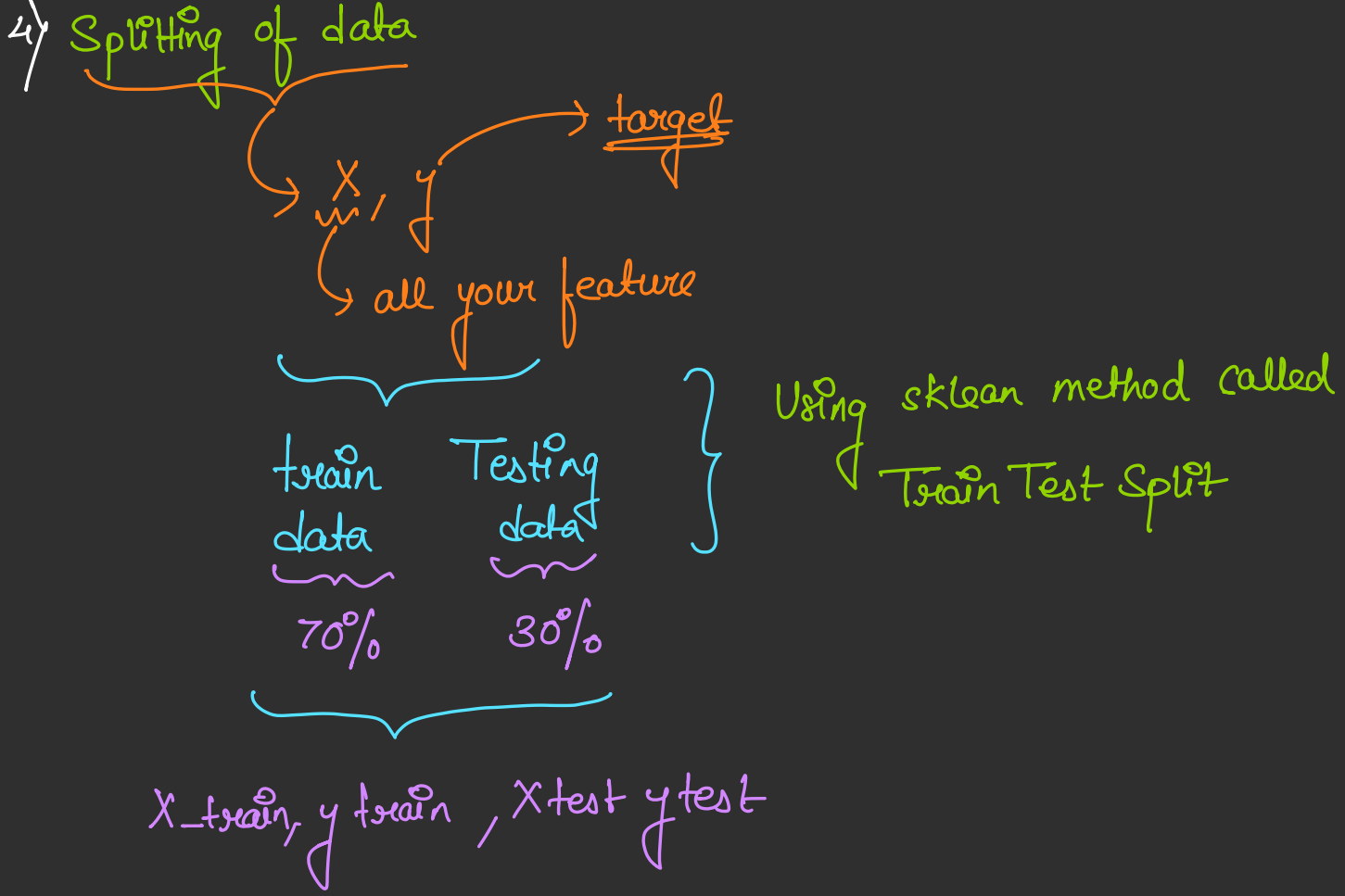
Target Encoding
Make, Model

Q) Why encoding is required?

→ Some ML model doesn't understand Text

Normalization / Std → Min-Max Scaler

→ Same scale


4) Splitting of data
target
X, y
all your feature
train data
Testing data
70%
30%
Using sklean method called
Train Test Split
X_train, y train , Xtest ytest

5) Model Training

1) Initialize the model ⇒ `reg = LinearRegression()`

2) Train the model by passing X_train & y_train ⇒ `reg.fit(X_train, y_train)`

Supervised learning

3) Testing the result ⇒ `reg.predict(X_test)`

↳ X_test, y_test

y_pred

6) Evaluation of Model

→ $y_{pred} == y_{test}$ ~~[$y_{hat} == y$]~~

~~R_2~~ → 0.945

→ Loss function

- MAE ⇒ Mean absoute error
- MSE ⇒ Mean Squared error

[min Loss Function]

M_1 M_2

21 31

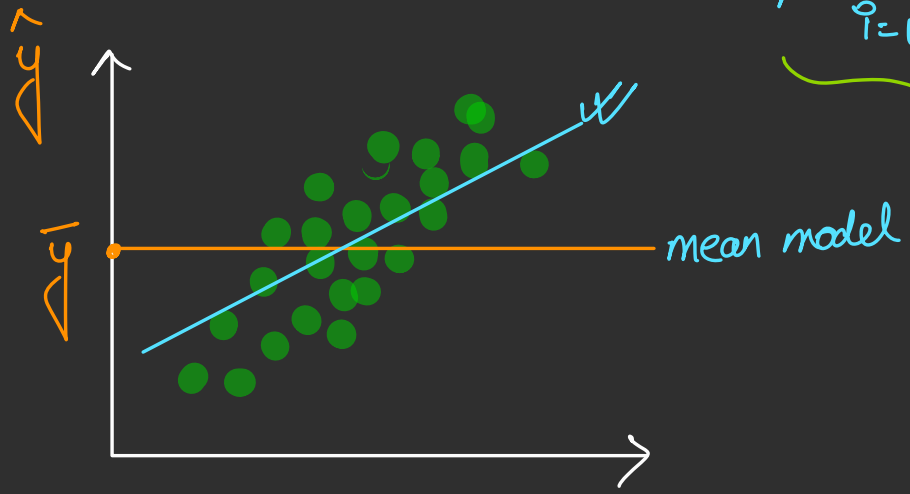
MSE

R_2 score = $1 - \frac{SS_m}{SS_T}$

R_2 → $(-\infty, 1]$

SS_T → (mean model)

$$R_2 \text{ Score} = 1 - \frac{SS_m}{SS_T} = 1 - \frac{\frac{1}{M}\sum_{i=1}^{n}(\hat{y} - y_i)^2}{\underbrace{\frac{1}{M}\sum_{i=1}^{n}(\bar{y} - y_i)^2}}$$



Model Interpretibility

↳ I want to understand what the model has understand about the data

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots + w_0$$

features

w_1 → coef-

$$\hat{y} = w_1 x_1 + w_0 \Rightarrow \hat{y} = (-0.22)(\text{km_driven}) + 0.11$$

Case I

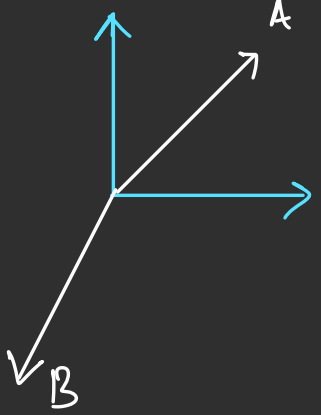
$\hat{y} = (-100)\,\text{age} + w_0 \;\Rightarrow\; \hat{y} = -100$

$\hat{y} = (-0.22)\,\text{km driven} + w_0 \;\Rightarrow\; \hat{y} = -0.22$

age $\longrightarrow$ 1 ↑

km driven $\longrightarrow$ 1 ↑

"Magnitude"



Case II

$$\hat{y} = (100)\,age + w_0 \longrightarrow \text{age}$$

$$\hat{y} = (0.32)\,kmd + w_0$$

Case III

$$\hat{y} = (100)\,age + (600)\,kmd + (0)\,color + w_0$$

weight is zero

Irrelevant feature

Quiz

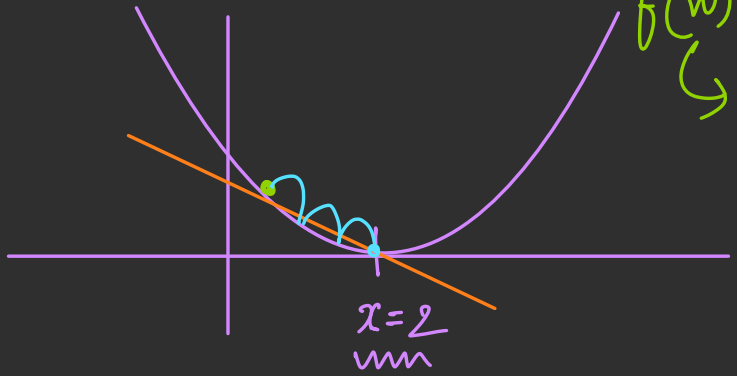
$$y = 5.2x_1 - 3.8x_2 + 2.1x_3 + 0.01x_4 - 1.5$$

good to drop this

(least Imp feature)

(accord to Model)

Gradient Descent



$f(w)$ = loss fᵉᵘⁿ

↳ weight fᵉᵘⁿ

Steps

1) Choose a random pt on $f(x)$

2) $\frac{df(x)}{dw}$

3) $w_{new} = w_{old} + \eta\left(\frac{-\partial f(x)}{\partial w}\right)$

Minimize our Loss function (MSE)

↳ Mean Squared Error

$$\underset{w, w_0}{\text{Minimize}} \quad \frac{1}{M} \sum_{i=0}^{m} (\hat{y}_i - y_i)^2$$

$$\hat{y}_i = \underbrace{w^T x + w_0}_{\text{vector form}}$$

Loss Function optimization

MAE

$$\frac{1}{m}\sum_{i=1}^{m}|y_i - \hat{y}|$$

MSE

$$\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y})^2$$

It is difficult

$$\frac{d\,MAE}{d\,y_{prod}} = \begin{cases} +1 & y_{ped} > y \\ -1 & y_{prd} < y \end{cases}$$

MSE

$$\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y})^2$$

$$L = \min_{w, w_0} \frac{1}{m}\sum_{i=1}^{m}\left[y_i - (w^T x + w_0)\right]^2$$

y_i : Actual

$w^T x + w_0$: pred

Example

$$L = \min_{w_1, w_0} \frac{1}{m} \sum_{i=1}^{m} \left[y_i - (w_1 x_1 + w_2 x_2 + w_0) \right]^2$$

$$\nabla L_{w, w_0} = \begin{cases} \frac{\partial L}{\partial w_0} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{cases}$$

$$L(w_1, w_2, w_0) = \{ y - (w_1 x_1 + w_2 x_2 + w_0) \}^2$$

$$\frac{\partial L}{\partial w_0} = \frac{\partial}{\partial w_0} \{ (y - w_1 x_1 - w_2 x_2 - w_0)^2 \}$$

$$\Rightarrow 2 (y - \underbrace{w_1 x_1 - w_2 x_2 - w_0}) \cdot (-1)$$

$$\Rightarrow -2 (y - \hat{y})$$

$$\frac{\partial L}{\partial w_2} = -2 (y - \hat{y}) \cdot x_2$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial}{\partial w_1} \{ (y - w_1 x_1 - w_2 x_2 - w_0)^2 \}$$

$$\Rightarrow 2 (y - w_1 x_1 - w_2 x_2 - w_0) \cdot (-x_1) \Rightarrow -2 (y - \hat{y}) x_1$$

Loss Function

↳ MSE

$$\frac{\partial L}{\partial w_0} = \frac{1}{m} \sum_{i=1}^{n} -2(y - \hat{y})$$

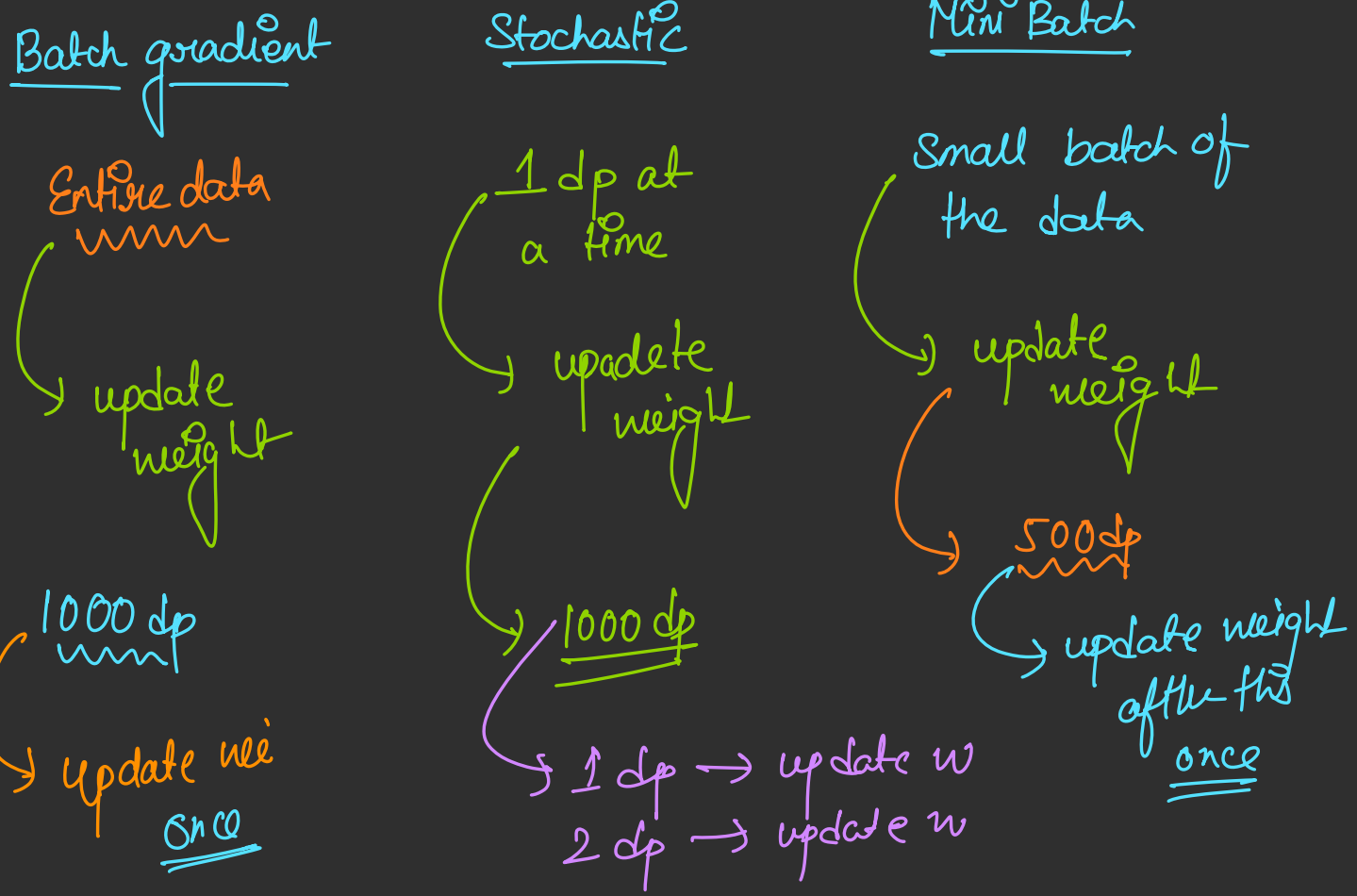
$$\frac{\partial L}{\partial w_1} = \frac{1}{m} \sum_{i=1}^{n} -2(y - \hat{y}) \cdot w_1$$

Calculating my GD

$$w' = w_{old} - \eta \frac{\partial L}{\partial w}$$

$$w_0' = w_{0old} - \eta \frac{\partial L}{\partial w_0}$$

The first value of w, w_0 will be be random choose


Batch gradient
Entire data
update weight
1000 dp
Update wei once
Stochastic
1 dp at a time
update weight
1000 dp
1 dp → update w
2 dp → update w
Mini Batch
Small batch of the data
update weight
500 dp
update weight after this once

Doubts

X_test, y_test
Not standardize yet

scaler = StandardScaler()
Normalize my data → Entire data training (X train, y train)

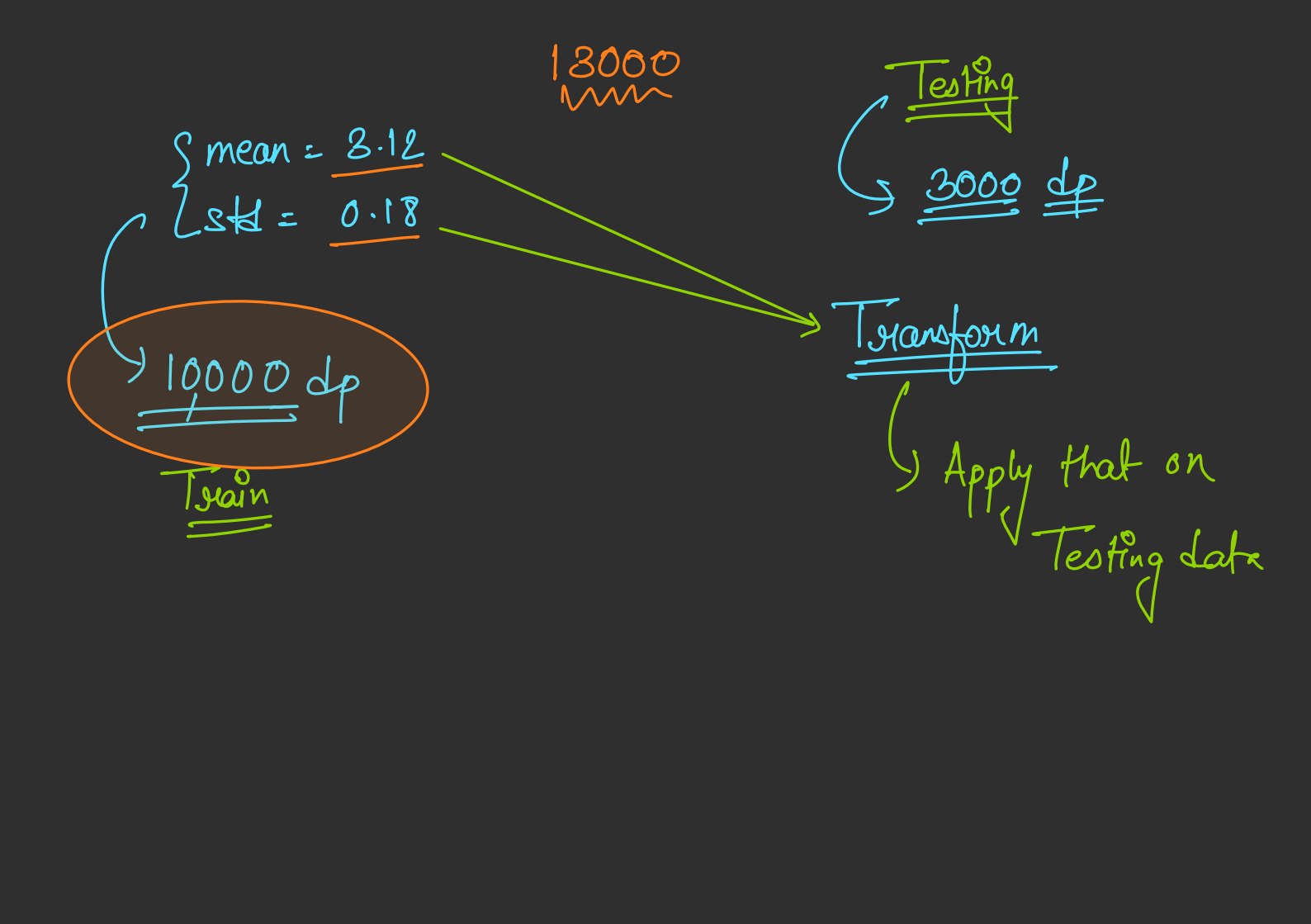
fit
fit_transform
Training data
calculate

$$Z\ score = \frac{x - x.mean}{std}$$

- scaler.fit
- scaler.transform

transform Only on test

scaler.transform(X_test)
scaler.transform(y_test)


13000
mean = 3.12
std = 0.18
10000 dp
Train
Testing
3000 dp
Transform
Apply that on
Testing data

$$\frac{\partial L}{\partial w_d} = -2(y - \hat{y}) \cdot x_d$$

pred $\Rightarrow \hat{y} = w^T x + w_0$

$(w^T x + w_0) \cdot y < 0$ (misclassified)

$w = w + x$

$w = w - x$

$y = +1$

$y = -1$

Actual values of y

$w = w + yx$

$y_i \cdot (w^T x + w_0) < 0$

$w = w - 2$